Sinclair Hudson *Machine Learning Engineer*

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Education

University of Toronto

Master of Science in Applied Computing, Artificial Intelligence Concentration *Sept '23 – Dec '24*

- 4.0 GPA with courses in natural language processing, software engineering, and advanced data systems.
- Vector Scholarship in Artificial Intelligence.

University of Waterloo

Bachelor of Computer Science, Honours, Co-operative Program, AI Specialization *Sept '18 – Apr '23*

- 93.50 major average, with courses in computer vision, robotics, and machine learning.

Relevant Work Experience

NVIDIA — San Francisco, CA

Machine Learning Engineer *Jan '25 – Present*

- Profiled and optimized a multitask perception model's training with Nsight Systems, increasing throughput by 32%, saving the org 60,000 H100 hours per week.
- Designed and built a performant dataloader for an internal data format, scaling training data 21× from 90k clips to 1.9M clips.
- Trained and tested end-to-end driving foundation models for on-car deployment, using a SLURM cluster to iterate on large-scale experiments.
- Cut over-budget chain-of-thought outputs by 99% for an on-car VLA driving model by training a custom tokenizer and normalizing training labels.
- Developed novel LLM-as-a-judge metrics to evaluate chain-of-thought reasoning for autonomous vehicle VLA models. Improved correlation with human judgment by 22% over existing metrics.
- Introduced a new parametrization for 3D cuboid orientation prediction, reducing roll error by 50% in a LiDAR object detection model.
- Integrated DINOv2 image features into the LiDAR object detection model using a novel camera-LiDAR attention mechanism.

NVIDIA — Santa Clara, CA

Deep Learning Researcher for Autonomous Vehicles *May '24 – Dec '24*

- Invented novel self-supervised learning techniques for LiDAR data in autonomous vehicles (patent pending).
- Built a PyTorch research framework for self-supervised pretraining on 2 million LiDAR scenes, implementing LiDAR variants of JEPA, VICReg, MAE, DINO, and world models.
- Ran over 100 pretraining and finetuning experiments evaluating transfer to LiDAR object detection, using custom metrics and visualizations to analyze learned representations.

Cruise — San Francisco, CA

Model Deployment Platform Engineer *Sept '22 – Dec '22*

- Designed and developed debugging tools to assist in deep learning model deployment, reducing deployment time by days in some cases.
- Built a tool to automatically assess numerical divergence between the original PyTorch models and optimized TensorRT models, allowing deployment engineers to quickly identify optimization errors.
- Built a linter to map ONNX nodes to lines in PyTorch source code, saving deployment engineers hours of manual debugging every deployment.
- Created a tool to manually add outputs to exported ONNX graphs, allowing engineers to inspect intermediate activations while debugging models.

NVIDIA — Santa Clara, CA (remote)

Deep Learning Researcher for Autonomous Vehicles *Jan '22 – Apr '22*

- Designed and iterated on multiple experiments for a LiDAR object detection neural network, improving cyclist and pedestrian detections by 43% and 15%, respectively.
- Implemented sparse tensor object detectors using Minkowski Engine, outperforming the production model while using 70% less memory.
- Integrated confidence predictions into a LiDAR object detection auto-labeling pipeline, allowing human annotators to focus efforts on anomalous and challenging data instances.

DarwinAI — Waterloo, ON (remote)

Machine Learning Developer *May '21 – Aug '21*

- Built and tested defect detection deep learning solutions for clients in the manufacturing industry, working in cross-functional teams and presenting results to stakeholders weekly.
- Trained XGBoost and SVR systems to model the relationship between environmental conditions and yield for an agriculture client, giving the client insight into how they can optimize their facility to maximize their yield.
- Implemented the core functionality of Dataset Distillation using the autograd package, to pursue research in low-data machine learning contexts.

- Reformulated defect detection as an unsupervised anomaly detection task, by implementing autoencoder experiments in PyTorch and validating them on client datasets.

Untether AI — Toronto, ON (remote)

Software Developer — *Sept '20 – Dec '20*

- Contributed to a customer-facing Python library to quantize TensorFlow computation graphs, towards compiling models for custom hardware.
- Designed and implemented Non-Max Suppression for quantized values using only integer operations, eliminating a CPU bottleneck and allowing object detection pipelines to run 10x faster on-chip.
- Experimented with different quantization schemes to improve object detection performance of an SSD-ResNet-34 model by 5%.

Huawei — Markham, ON

LiDAR Perception Researcher — *Jan '20 – Apr '20*

- Implemented key modules from 12 different research papers in PyTorch, summarizing state-of-the-art techniques and enabling further research.
- Analysed the SemanticKITTI dataset to produce optimal class loss weights, boosting mIoU by 2%.
- Wrote a data loader to spatially align sequential LiDAR scans for temporal pipelines, based on IMU data, enabling the research of multi-scan LiDAR segmentation pipelines.

Peer-Reviewed Publications

Sim-to-Real Domain Adaptation for Lane Detection and Classification in Autonomous Driving — *IV 2022*

Application of Machine Learning for Drone Classification Using Radars — *SPIE Defense + Security 2021*

Skills

Topics: Self-Supervised Learning, Computer Vision, Large Language Models, Foundation Models

Languages: Python, C++, R, C, LaTeX

Frameworks: PyTorch, NumPy, Pandas, TensorFlow, OpenCV, ROS, Gradio

Tools: Git, Docker, Conda, Bazel, CARLA, Linux, TensorRT, Weights and Biases, SLURM